

Chapter 4

1.

- a) `If (grade >= 90) {
 alert(" Great job!");
}`
- b) `If (number < 20 || number > 50) {
 alert("error");
}`
- c) `If (y < 100) {
 Y += 2;
}`

```
2. If (num1 > num2) {  
    alert ("First number is larger.");  
}  
Else if (num2 > num1) {  
    alert("second number is larger.");  
}  
Else {  
    alert("Numbers are equal.");  
}
```

3,. Even because:

If `(num % 2 == 0)` checks if the number has a remainder of 0 when divided by 2

- 4. A) `random.nextInt(50) + 1`
- b) `random.nextInt(81) + 20`
- c) `random.nextDouble() * 10 + 10`

5. `Age == 18` is not included, `age == 65` is not included.

6.

- a) true
- b) False
- c) true
- d) true
- e) true
- f) True
- g) false

8.

- a) true
- b) false
- c) false
- d) false
- e) true
- f) false
- g) true
- h) Ture
- i) false
- j) ture
- k) true
- l) true